HISTORICAL INTELLIGENCE REPORT

Prometheus Grid Intelligence | 2015-12-31T00:00:00 to 2021-12-31T00:00:00

MODEL PERFORMANCE COMPARISON

Model	MAE (MWh)	RMSE (MWh)	MAPE (%)	Status
Synthetic Baseline	2276.45	2914.81	19.48	REVIEW
LightGBM Quantile Ensemble	2276.45	2914.81	19.48	REVIEW
Adani LightGBM	2276.45	2914.81	19.48	REVIEW

ANOMALIES DETECTED (0)

No significant anomalies detected in the selected period.

OPERATIONAL INSIGHTS

During this period, the **Synthetic Baseline** model demonstrated the highest accuracy with a MAPE of 19.48%. High forecasting precision directly translates to optimized grid stability and reduced operational costs by minimizing emergency power procurement.

Disclaimer: This report is an automated diagnostic tool for grid officers. Please cross-reference with SCADA real-time telemetry before critical switching operations.

INSTITUTIONAL COMPENDIUM

AGEL / AESL benchmarked financial intelligence layer

PORTFOLIO HEALTH: ACTUALS VS ADANI BENCHMARKS

Indicator	Actual	Benchmark	Status
Realized Tariff (■/kWh)	3.263	3.09	DRIFTING
Unit O&M Cost (■/kWh)	0.450	0.45	ON TARGET
Transmission Availability (%)	99.60	99.6	INCENTIVE
Receivable Cycle (DSO days)	44.7	23	STRESSED
Debt / EBITDA (x)	2.00	2.2	STABLE

SUSTAINABILITY IMPACT

Net energy billed: **7910033.35 MWh** | Thermal replacement factor: **0.8 kg CO2/kWh** | CO2 emissions avoided: **6328026.68 tonnes**.

FINANCIAL WATERFALL

Step	Amount (■ Crore)
Gross Revenue	2580.72
Less: Operating Expense	-375.83
Operational Cash Profit (EBITDA)	2204.89
Add: Carbon Credit Potential	158.20

Benchmark notes: Incentive status is active when availability stays above 99.6%. Financial health aligns with ~23-day DSO discipline. Investment-grade watch is triggered if debt-to-EBITDA exceeds 2.2x or EBITDA weakens materially.